

*DataSoc x Atlassian Datathon 2025*

# From Risk to Growth: Forecasting Atlassian Customers



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# Problem Statement

For Atlassian, **customer retention** and **expansion** are critical to sustainable growth and maximising lifetime value.

But how do we know in advance whether a customer is likely to upgrade, downgrade or expand their seats – and what's driving those decisions?

↑ UPGRADE ↑

↓ DOWNGRADE ↓

↔ EXPAND ↔

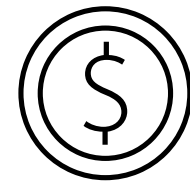
*Anticipate customer moves – not react after they happen.*

# Why Should Atlassian Care?



## Business Stake

- ⚠ Retention and expansion drive Atlassian's growth



## Cost of Inaction

- ✗ Churn
- ✗ Lost revenue
- ✗ Disengaged users



## Opportunity

- ✓ Predict customer moves
- ✓ Intervene early
- ✓ Maximise lifetime value
- ✓ Unlock growth

# 55% of Atlassian users who change plans **downgrade**

*How can we change these numbers?*

## Model 01

- Predict whether a customer will **upgrade** or **downgrade**
- Logistic Regression
- **Key Drivers:** Discounts, seats, support tickets, actions

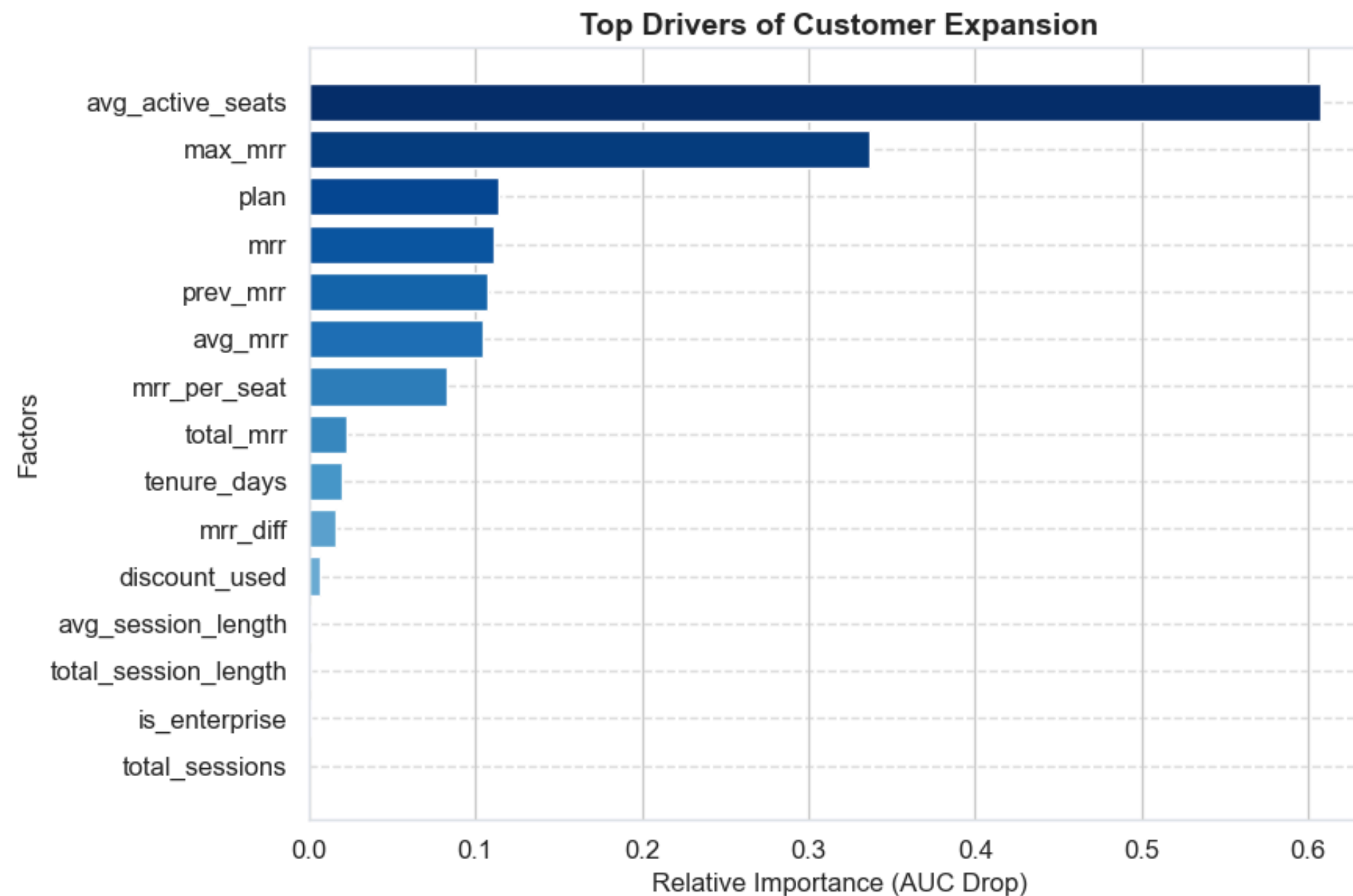
*Identify **at-risk** accounts early → support before they leave*

## Model 02

- Predict whether a customer will **expand** (increase active seats)
- Neural Network
- **Key Features:** Usage history, engagement signals, account growth trends, seats

*Identify **expansion-ready** accounts → capture growth opportunities*

# What Are The Drivers For Expansion?



## Upsell priority:

- Expansion is driven most by active seats and revenue (MRR levels)
- **Larger, higher-spending** accounts are the strongest signals

## Lower impact:

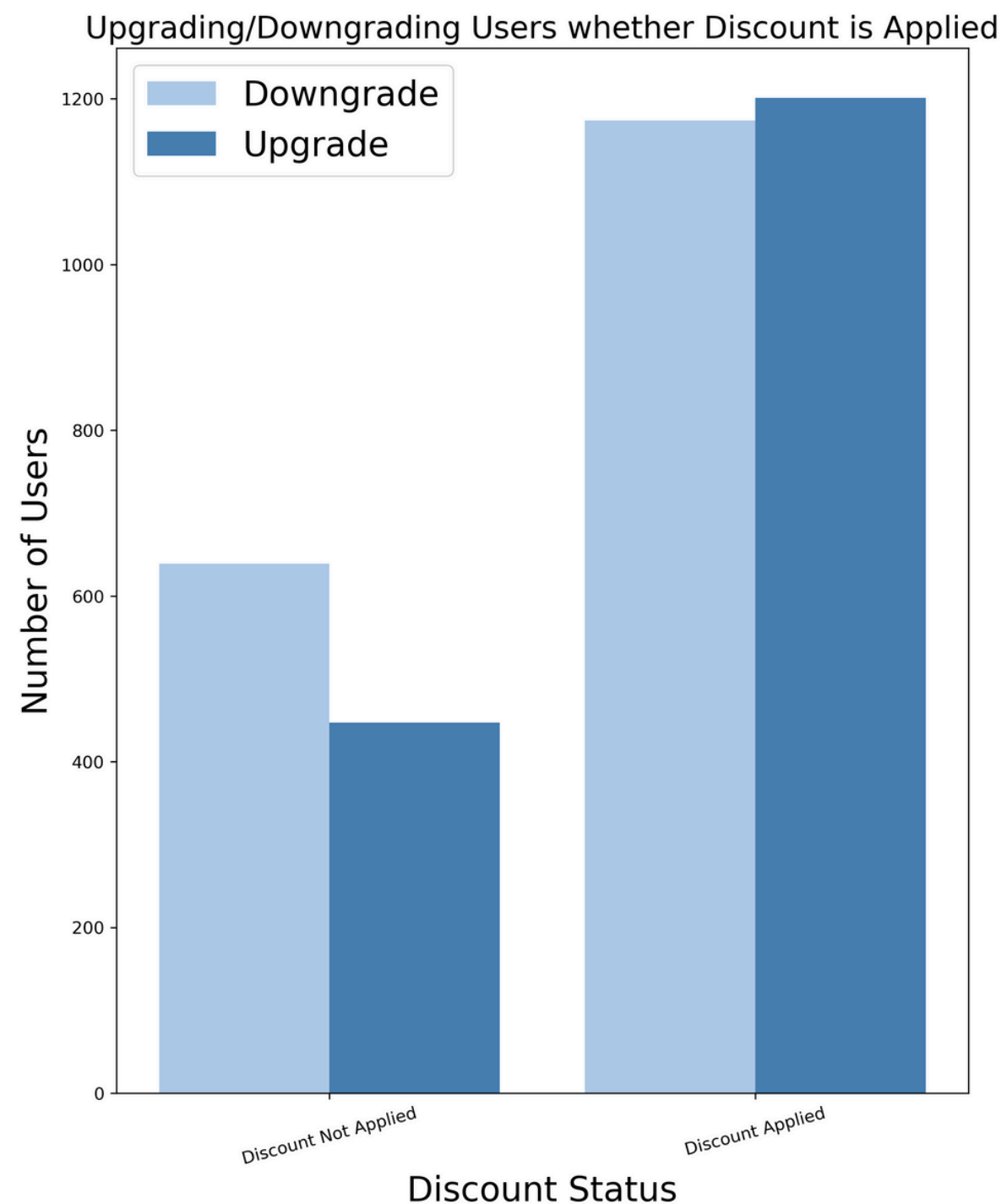
- Tenure, discounts, session usage and support activity add little predictive power in this dataset

# Model Results

| <u>Model</u>               | <u>Accuracy</u> | <u>Precision</u> | <u>Recall</u> | <u>F1</u> | <u>AUC</u> |
|----------------------------|-----------------|------------------|---------------|-----------|------------|
| <u>Maximising Accuracy</u> | 87%             | 96%              | 87%           | 91%       | 95%        |

- Our model achieves 96% precision → when we predict an upgrade, we're almost always right
- 87% recall means we capture most upgrade opportunities, though a few slip through
- **Business impact:** Sales can confidently target flagged accounts, with minimal wasted effort

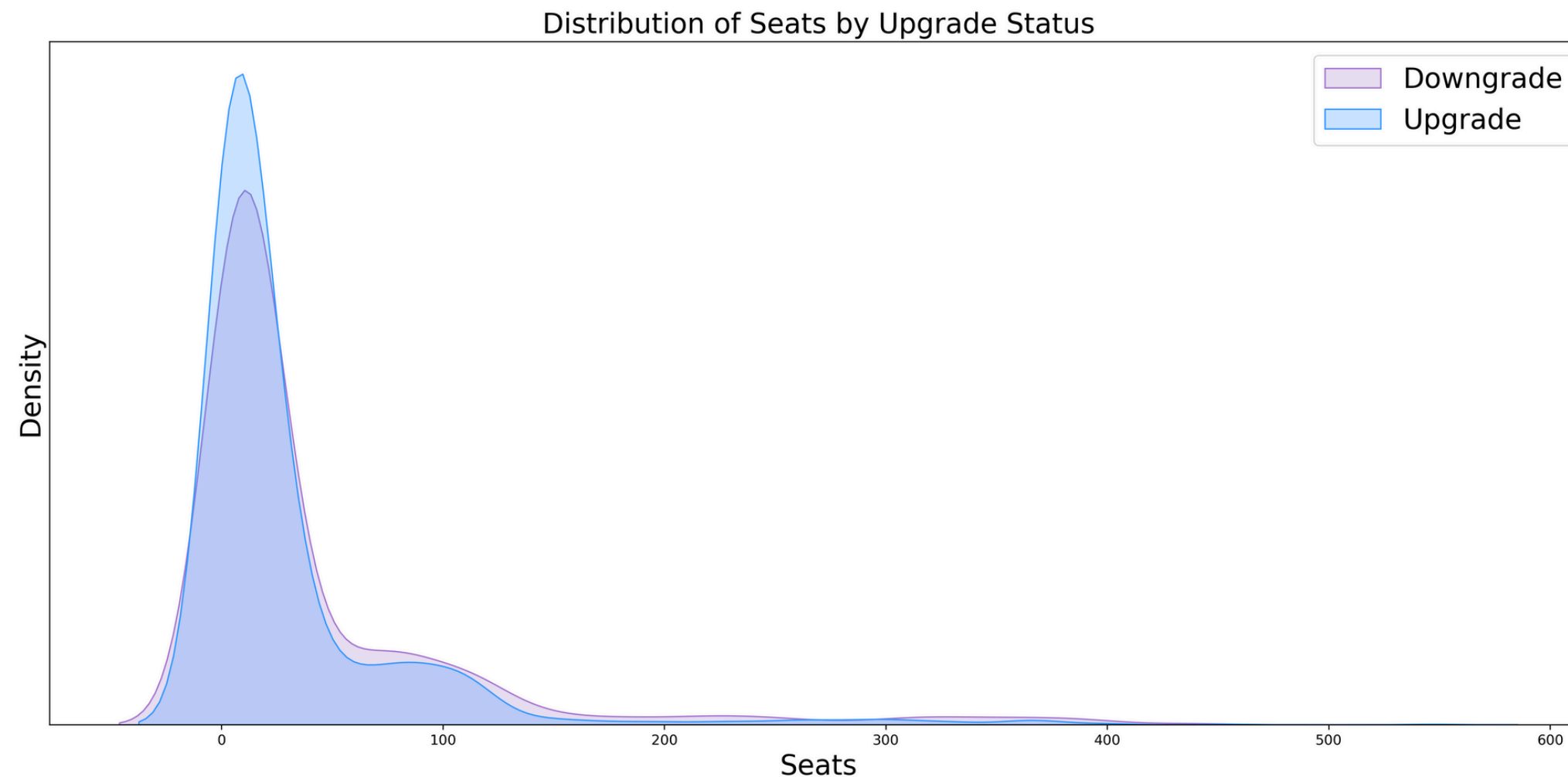
# What Are The Drivers For Upgrades?



## Discounts

- Discounts **increase** the proportion of users that upgrade
- **Why?** Lower barrier to entry, perceived value boost, budget fit

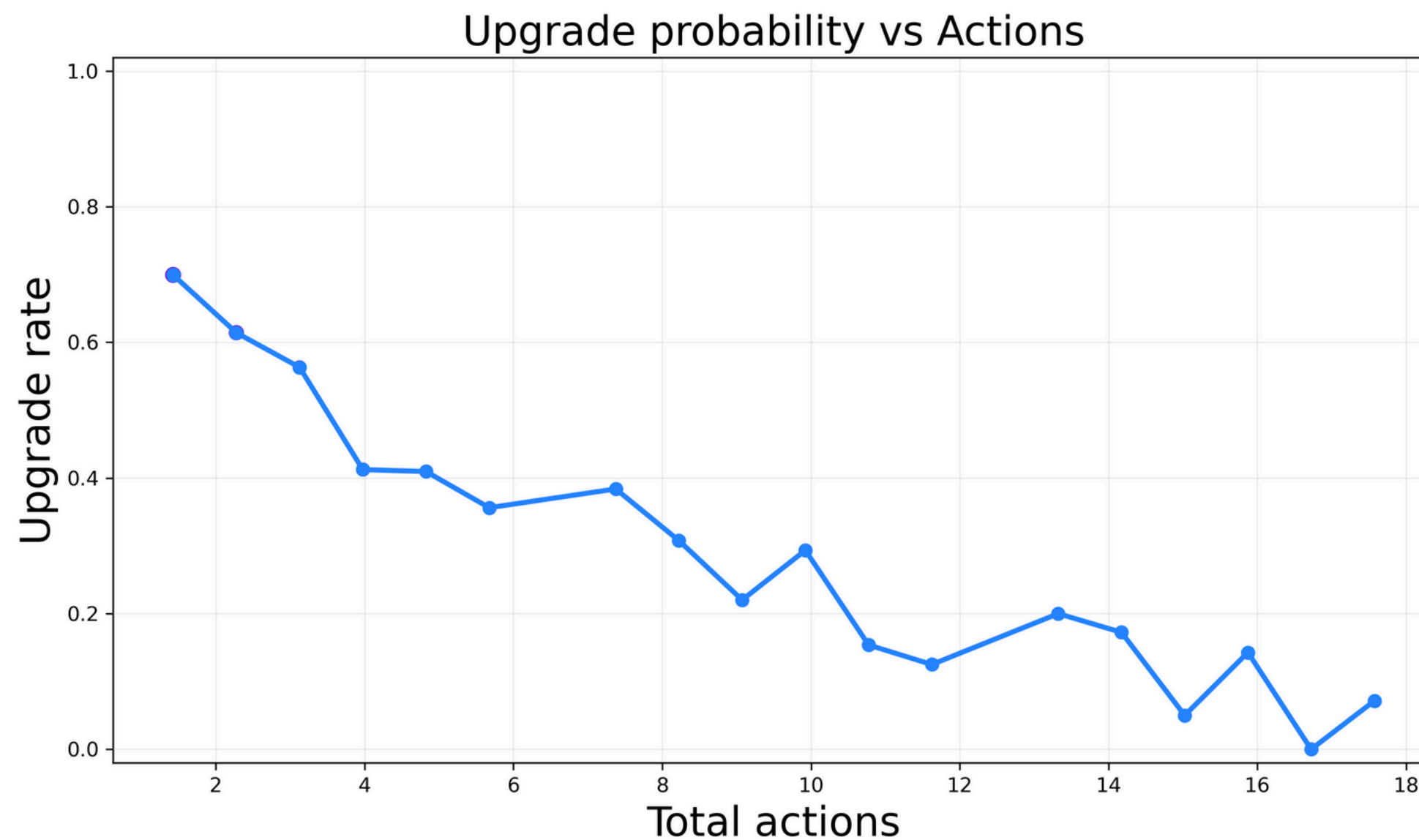
# What Are The Drivers For Upgrades?



## Seats

- Increasing number of seats will **increase** proportion of upgrading users (until ~25 seats)
- Larger accounts are most likely to both **expand** and **upgrade**
- **Why?** Greater scale → capacity limits, more teams, more opportunities

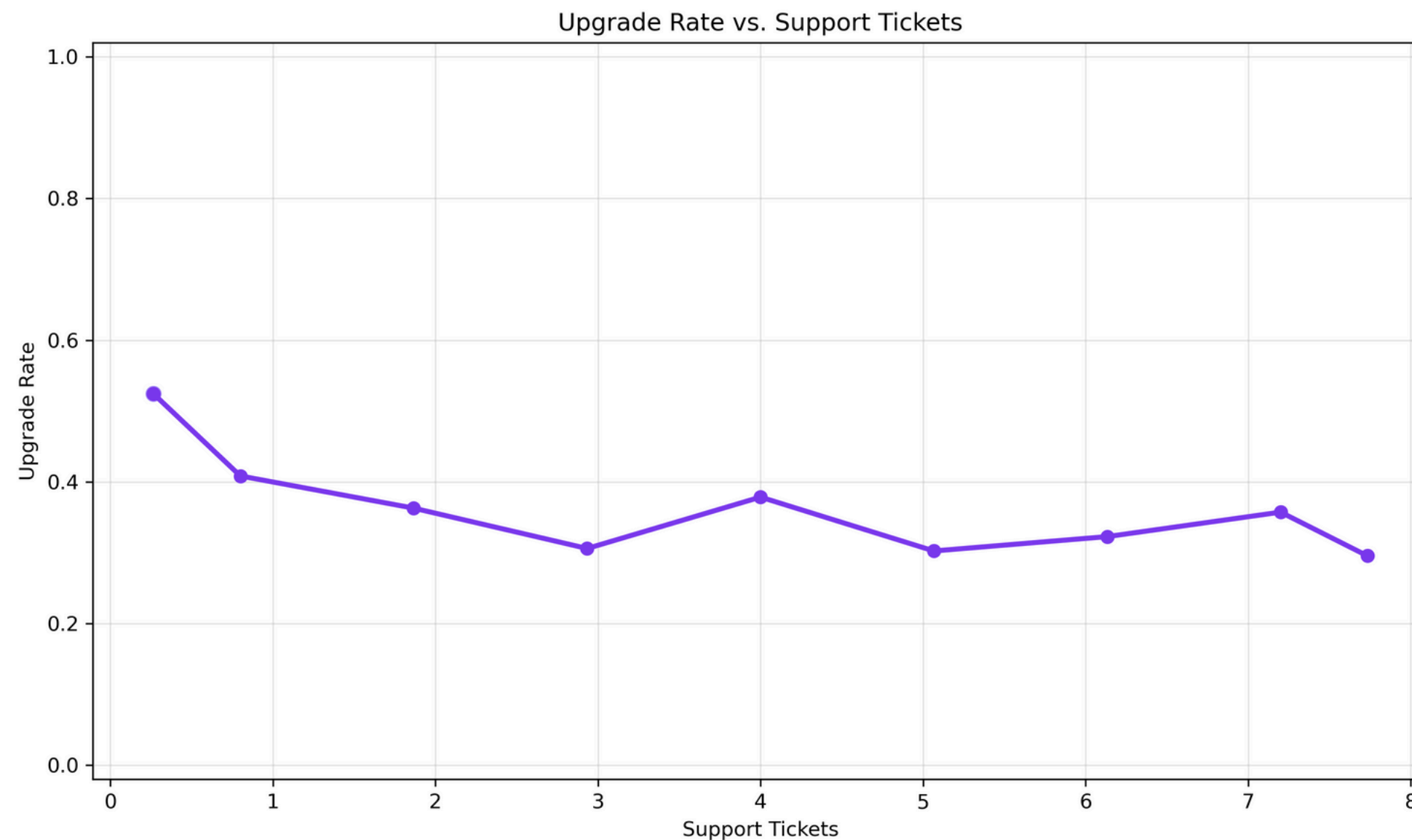
# What Are The Drivers For Downgrades?



## Actions

- Surprisingly, **more actions** taken on an Atlassian software **decreases** the upgrade rate
- **Why?** High activity may reflect struggling users or friction from heavy reliance without satisfaction

# What Are The Drivers For Downgrades?



## Support Tickets

- Increasing the number of support tickets sent **decreases** upgrade rate
- **Why?** High ticket volume signals frustration, friction or unresolved product issues

# Scenarios:

## *Predicted vs. Actual Customer Behaviour*

|                              | <u>Predicted: Downgrade (1)</u>                                                                                                                                                                   | <u>Predicted: Upgrade (0)</u>                                                                                                                                                                                       |
|------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <u>Actual: Downgrade (1)</u> | <p><b>True Positive (TP)</b><br/>Customer is truly at risk of downgrading and the model predicts downgrade.</p> <p><b>Business Consequence:</b><br/>Discount is offered, improving retention.</p> | <p><b>False Negative (FN)</b><br/>Customer is truly at risk of downgrading but the model predicts upgrade.</p> <p><b>Business Consequence:</b><br/>No discount is offered → customer downgrades → revenue loss.</p> |
| <u>Actual: Upgrade (0)</u>   | <p><b>False Positive (FP)</b><br/>Customer is actually upgrading, but the model predicts downgrade.</p> <p><b>Business Consequence:</b><br/>Discount is given unnecessarily → lower margins.</p>  | <p><b>True Negative (TN)</b><br/>Customer is truly upgrading and the model predicts upgrade.</p> <p><b>Business Consequence:</b><br/>No discount is offered, customer upgrades as expected.</p>                     |

# Model Selection



RandomForest



XGBoost



GradientBoost



Logistic Regression

| Monitor              | Recall Metric |
|----------------------|---------------|
| LogisticRegressionCV | 0.8953        |
| GradientBoosting     | 0.7741        |
| XGBoost              | 0.7201        |
| RandomForest         | 0.7131        |

**Minimizing: False Negative (FN)**  
 Customer is truly at risk of downgrading  
 but the model predicts upgrade.

**Business Consequence:**  
 No discount is offered → customer  
 downgrades → revenue loss.

- LogisticRegression has highest Recall Metric of **0.8953**.
  - Successfully flags almost 9 out of 10 customers that would downgrade
- The model **prioritises customer retention** over discount efficiency
- We also choose Logistic Regression for its **interpretability** and **simplicity**

# Strategies for Retention:

## *Protect Current Revenue*

1

### High ticket accounts = early intervention

- **Monitor** support tickets as a leading churn indicator
- Prioritise accounts with repeated/severe issues for **proactive outreach**
- **Personalised solutions** and **early support** outreach to prevent frustration

2

### Targeted incentives

- For customers at **downgrade risk**, targeted **discounts** can extend lifetime value
- Use discounts **selectively**: keep engaged accounts without devaluing pricing
- Pair discounts with **engagement drivers** such as training, feature adoption and integrations

# Strategies for Expansion:

*Unlock Growth*

1

## AI-Flagged Growth Signals

- Highlight accounts showing strong product adoption, increased seat utilisation or new team onboarding

2

## Pre-Capacity Outreach (Proactive Engagement)

- Direct sales toward accounts at ~70–80% seat utilisation, before they reach capacity

3

## Expansion Triggers

- Automated playbooks activate when leading signals emerge
- System-generated tasks ensure proactive sales motion and prevent missed expansion window

# Introducing... CRUSHR App



[Click to Access  
CRUSHR App](#)

- Tool for Atlassian to get personalised discount recommendations for each user profile!
- Calculates probabilities of upgrade/downgrade with and without discounts!